

2024

Table 1:

2024.09-2027.09
2021.09-2024.09

HgCdTe APD -

Sci. China: Phys. Mech. Astron. Infrared Phys. Technol. Infrared Laser Eng. / 3 Nat. Photonics IEEE J. Quantum Electron. Advanced Optical Materials SCI 5 IEEE Trans. Electron Devices Infrared Phys. Technol. APD Nat. Comput. Sci.

2017.07-2021.09

:

APD

- HgCdTe

MC² —

C++ - MPI/OpenMP (1) (2) (3)
(4) SQLite HgCdTe APD

MATLAB

C/C++ Python	MATLAB	SQLite	C++	MPI OpenMP
LabVIEW				
CET6				

2024.06

2024.06

2023.06

2022.06

2019.06

2019.06

2018.06

JW ZFB	XXX	363
		800
		20

Earth & Space from Infrared to Terahertz International Conference 2023.09.

”HgCdTe e-APD Microscopic-Mesoscopic Monte Carlo Paradigm for Excess Noise Analysis”

1. **Liu Shuning**, Xie Runzhang, Kang Qianlong, Ge Haonan, Luo Min, Liu Anna, Hao Xie, Chen Lu, Liu Xiwen, Li Qing, Hu Weida. “Mesoscopic behavior of single-carrier avalanche in narrow-bandgap semiconductors”. *IEEE Transactions on Electron Devices*, **73**(4), 2038-2044 (2026).
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3. Xie Runzhang #, **Liu Shuning** #, Ge Haonan, Li Qing, Dai Fuxing, Zhang Tao, Liu Anna, Luo Min, Gao Wei, Wang Peng, Zhang Zhenhan, Wang Zhen, Xu Hangyu, Wang Hailu, Fu Xiao, Wang Fang, Miao Jinshui, Hu Weida. “Allowing sparse sampling of wavefield toward phase driven optics and artificial intelligence”. *Nature Photonics*, (Under review, 2024). (#)
4. **Liu Shuning**, Han Qi, Luo Wenjin, Lei Wen, Zhao Jun, Wang Jun, Jiang Yadong, Markus B. Raschke. “Recent progress of innovative infrared avalanche photodetectors”. *Infrared Physics & Technology*, **137**, 102114 (2024).
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7. Luo Min #, Ge Haonan #, Li Qing, Gao Wei, Kang Qianlong, Liu Anna, **Liu Shuning**, Xie Runzhang, Wang Peng, Hu Weida. “Sub-wavelength optical modulation of step-size varying infrared photodetectors”. *Infrared Physics & Technology*, **145**, 105609 (2025). (#)
8. Wang Hao #, **Liu Shuning** #, Wang Peng, Ge Haonan, Chen Yue, Wang Hailu, Xu Tengfei, Guo Jiaxiang, Zou Yuanchen, Wei Wenrui, Jiang Ruiqi, Wang Fang, Martyniuk Piotr, Miao Jinshui, Hu Weida. “High-performance violet phosphorus photodetectors with van der Waals-assisted contacts”. *Science China Physics, Mechanics & Astronomy*, **9**, 297311 (2023). (#)
9. Gu Yue, Xie Runzhang, Wang Peng, Li Qing, Zou Yuanchen, **Liu Shuning**, Zhang Kun, Wang Yang, Wang Hailu, Wang Fang, Chen Lu, Hu Weida. “Bandwidth Characteristics of Mid-Wavelength Infrared PIN HgCdTe Avalanche Photodiodes”. *IEEE Journal of Quantum Electronics*, **59**, 1 (2023).
10. Li Qing, Xie Runzhang, Wang Fang, **Liu Shuning**, Zhang Kun, Zhang Tao, Gu Yue, Guo Jiaxiang, He Ting, Wang Yang, Wang Peng, Wei Yanfeng, Hu Weida. “SRH suppressed P-G-I design for very long-wavelength infrared HgCdTe photodiodes”. *Optics Express*, **30**, 16509 (2022).
11. **Liu Shuning**, Tang Qianying, Li Qing. “Research progress on local field characterization of mercury cadmium telluride infrared photodetectors (invited)”. *Infrared and Laser Engineering*, **51**, 20220277 (2022). (EI)

12. Li Qing, **Liu Shuning**, Wang Fang, Tang Qianying, Guo Jiaxiang, Hu Weida. “Light-Triggered and Polarity-Switchable Homojunctions for Optoelectronic Logic Devices”. *Advanced Optical Materials*, **11**, 2202379 (2023).

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